

Dalal Ratings: Deep Learning Based Platform for Stock Investors

Harjaspreet Singh Wadhwa
ECED, Thapar Institute of
Engineering and Technology Patiala,
India
hwadhwa1_be20@thapar.edu

Jasmeet Wadhwa
Director Engineering
GlobalLogic India Pvt Ltd
Noida, India
jazmeetwadhwa@gmail.com

Kanwarpreet Kaur
ECED, Thapar Institute of
Engineering and Technology Patiala,
India
kpreet2392@gmail.com

Neeru Jindal
ECED, Thapar Institute of
Engineering and Technology
Patiala, India
neeru.jindal@thapar.edu

Kulbir Singh
ECED, Thapar Institute of
Engineering and Technology
Patiala, India
ksingh@thapar.edu

Abstract—It is quite difficult to analyze the stocks and make investments in the stock market especially by the small investors. Therefore, it is required to develop a platform for investors to compare stock analyzers/brokers using the real-time rating system. The Dalal rating portal is developed which makes it easy for investors to use the real-time rating system and select whose recommendation to follow. All the data is authentic and is updated daily from the official National Stock Exchange (NSE) website. Further, Long Short-Term Memory (LSTM) Neural Networks are employed for predicting the stock market. The existing platforms provide collective recommendations but cannot verify the same. Hence, the Dalal Rating portal becomes a unique one-stop platform that introduces an automatic broker rating system and also the feature to verify recommendations that are not present in the existing platforms. This portal can filter collective recommendations, access the latest news, perform stock forecasting, and a detailed view of each analyzer, thus, improving the user experience.

Index Terms—Stock Market, National Stock Exchange (NSE), Long Short-Term Memory (LSTM) Neural Networks, Stock Analyst, Stock Broker

I. INTRODUCTION

Share Market Equity Market or Stock Market are all synonyms of one another. Stock Market is a legal platform for trading of equity/shares of publicly registered organizations. The stock market can be divided into primary and secondary market. Here, primary market is the one where the company sells its shares to investors (general public) and the secondary market is that where investors buy and sell shares among themselves. Such markets operate under specific regulations. In India, SEBI governs this process [1]. India's top Stock Exchanges are National Stock Exchange (NSE) and Bombay Stock Exchange (BSE). Before a company can offer its shares to the public, it needs to register with a stock exchange, such as NSE or BSE. These stock exchanges also have registered people with great knowledge of the stock market and are experts in stock analysis; these professionals are called Stock Brokers. Stock Brokers act as middlemen in guiding new

investors who have just entered the stock market to make the right decisions and effectively invest their money. Stock brokers are the first choice for almost everyone investing in the share market. Brokers are licensed to perform the Buying and Selling of shares on behalf of the respective investor [2]–[7]. Fig. 1 represents the workflow of the Stock Market in India.

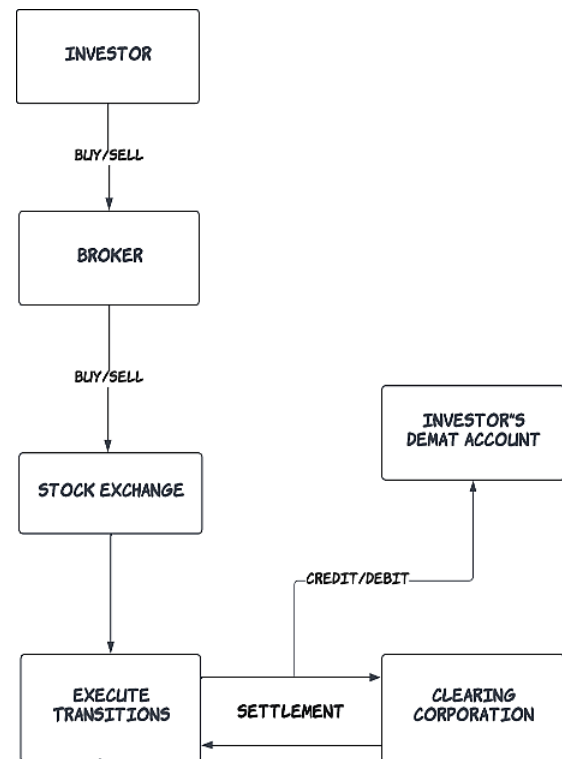


Fig. 1. Workflow of Stock Market

Stock brokers study the stock market inside out and, based on the latest news updates, give their opinions on the price movements of various stocks. Based on these opinions, new

and less experienced investors take their chances and invest their hard-earned money. These opinions by Stock Brokers are based on thorough research and are called Stock Broker Recommendations [5], [8].

Stock Analysts predict the flow of the market using these techniques and share recommendations publicly. These analysts make use of the latest tools, many market indicators, and various research techniques for their predictions [9]. To effectively estimate these markets, many researchers and companies are continually researching and developing various state-of-the-art approaches and algorithms to give quantitative predictions [10]–[12]. Nevertheless, analysts remain to be one of the most important sources of information. Therefore, they should serve as one of your main starting points for determining whether or not the company is trading at a premium or a discount. They should not, however, be your only source of opinion [5].

There exist web-based platforms that provide news and collective recommendations, but they are unable to provide verified recommendations. Further, there is no way of classifying a Broker as Good or Bad. Thus, people who don't have the skill of analyzing the market don't know whom they take advice before making investments. Even, there is no automatic broker rating system. Hence, a platform is introduced in the proposed work that provides a solution to these problems. The contributions of the proposed work (Dalal rating portal) are as follows:

- To provide the real-time broker ratings.
- One-stop Trusted platform for all Stock Market professionals that provides unbiased automated broker rating system.
- We are making the Analyst/Broker accountable by saving all recommendations and providing a Rating.
- Collective Stock Recommendations by all Brokers and a User-friendly interface to get information on a specific Broker/Stock. Thus, making it easy for investors to compare various Analysts/Brokers.

II. RELATED WORKS

Stock Market dramatically impacts the economic development of a country's financial system, making it more important to have a trusted and authentic platform for all stock market professionals [13]–[18]. Here is a comparison between our proposed work and some existing technologies. In this research paper, we have used the terms stock analyst and stockbroker multiple times; here, we consider them to be the stock market professionals making recommendations. The related works are illustrated in Table I.

III. PROPOSED METHODOLOGY

In the proposed work, the portal is populated with recommendations from brokers/analysts either by themselves or by our Dalal agents. To authenticate the data, authorised personnel can submit recommendations only with a username and password. Then, the portal verifies and calculates the target date and finally displays the newly added record on the portal,

TABLE I
RELATED WORK

Platforms	Money Control [15]	Economic Times [14]	NSE [16]
Type	Online business publication	Business newspaper	Official website
Coverage	Latest news and views on business, markets, personal finance, mutual funds, investment ideas, and other aspects of the financial industry	Print editions in almost all metros and cities in India, online edition, provides business news via a website and mobile application	Provides investor-friendly information which is a trusted source for all the stock market action, includes list of all the registered companies and brokers in the stock market
Features	Curated markets data, actionable investment ideas, independent equity research, global and local news, opinions, views, and analysis	Curated markets data, e-learning, and equity research	Multi-tiered risk management system and equity market

visible to everyone. Every day at 3:30 PM, after trading hours, the portal evaluates the documents based on the status. If the status is MET or NOT MET, then no changes are made to the records. If the status is blank "-" then the complete record is updated with the real-time value of LTP, HIGH, and LOW, and the status is updated depending on the target date. If the target date is reached but the target price is not, then the status is updated to "NOT MET", and the met date is left blank. If the target price is reached before or on the target date, then the status is updated to "MET", and the met date is updated to the current date. As the status is updated, the rating of the respective analyst/broker also changes. Fig. 2 explains all the options available for navigating to a user visiting our proposed work website.

IV. DESIGN AND IMPLEMENTATION

This platform used the Python language Django framework and hosted on a virtual machine (VM) using Amazon Web Services (AWS). This work was thoroughly tested with real data for three months in Linux and Windows Operating Systems (OS). Some libraries used, and their utility is mentioned below:

- Pandas: This is used to access .csv files and then process them to get results.
- BeautifulSoup: For scraping real-time prices of different stocks from the screener.in
- Django Crontab: For scheduling the task of processing data every day at 3:30 PM
- Django Auth: For authenticating and authorizing user experience
- DateTime: This automatically calculates the target date and compares data in CSV files.
- smtplib: For email sending functionality

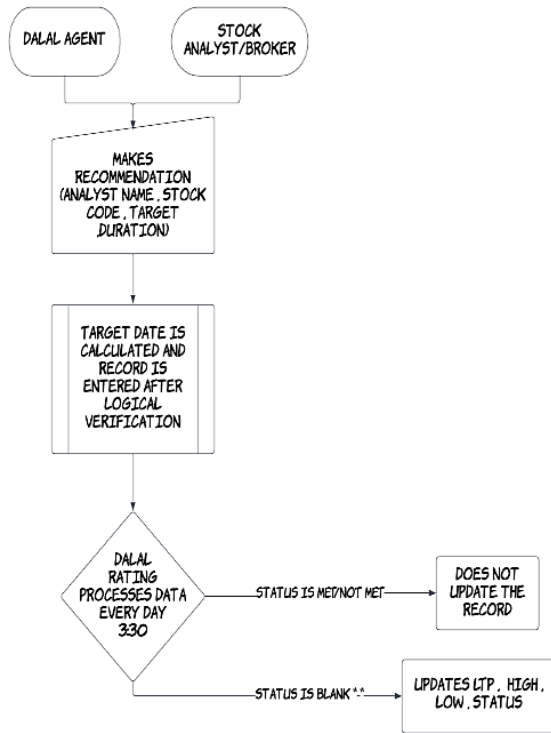


Fig. 2. Proposed Methodology

- Tensorflow/Keras: For Long Short-Term Memory (LSTM) Neural Network
- Scikit Learn: For data normalization and test-train split
- matplotlib: For plotting and visualizing data

The user navigation is illustrated in Fig. 3, which sums up and explains all the features of our proposed work for the user.

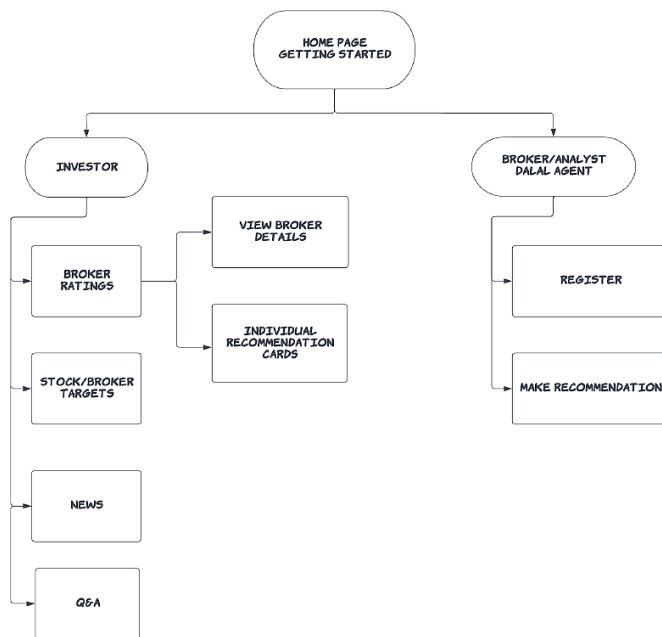


Fig. 3. User Navigation

Features of Dalal Ratings solution

(a) Home page This gives a basic Introduction to the Dalal Ratings Website, telling the user the benefits of using this platform and instructions for getting started, as shown in Fig. 4.

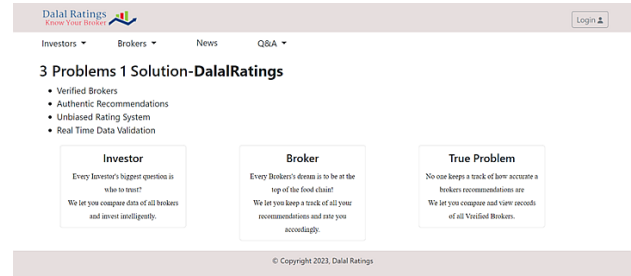


Fig. 4. Proposed work's (Dalal Rating portal) Home Page

(b) Broker Ratings Users can navigate to the page in Fig. 5(a) by clicking on Investors>Broker Ratings. This is the unique feature of our solution; ratings and information on registered verified brokers are displayed on this page. From this page, users can compare different Brokers based on their ratings and view all their recommendations in detail by clicking on "See Recommendations", which leads to the following personal page for each Broker, as given in Fig. 5(b).

(c) Filtering by Stock Users can navigate to the page in Fig. 5(c) by clicking on Investors>Stock Targets. This page lets the User view and filter stock recommendations by respective STOCK CODE. Fig. 5(d) shows Search Results for Stock Code INFY>.

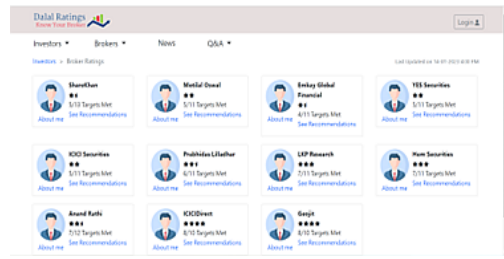
(d) Filtering by Broker/Analyst Users can navigate the page in Fig. 5(e) by clicking on Investors>Broker Targets. This page lets the user view and filter stock recommendations by respective BROKER NAME. Fig. 5(f) shows search results for Broker ShareKhan.

(e) Registration Users can navigate the page in Fig. 6(a) by clicking on Brokers>Register. Our website supports Registration by both Brokers and Users.

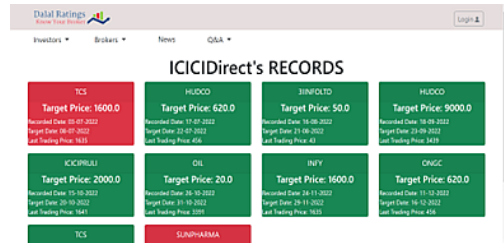
- **Registration for Brokers:** The user can navigate to this page by clicking on Broker Button. After correctly reading the instructions, the Broker can fill out the registration form shown in Fig. 6(b).

- **Registration for Dalal Agent:** This page shown in Fig. 6(c) opens by default after clicking on Broker>Register, or the user can navigate to this page by clicking on the DalalAgent Button. After correctly reading the given instructions, the Broker can proceed to fill out the registration form shown in Fig. 6(d).

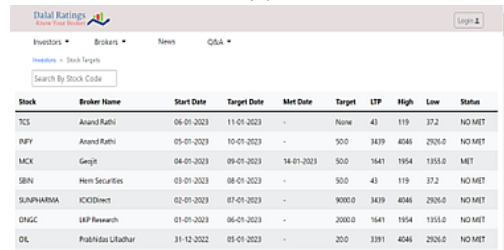
(f) Verified Recommendations The user can navigate the page in Fig. 7(a) by clicking on Brokers>Make Recommendations. This feature is limited to only DalalAgents (Verified people hired to upload data manually). Fig. 7(b) shows the make recommendation page view. Each DalalAgent is a verified person who can only upload recommendations of other Brokers that are publicly available from authentic



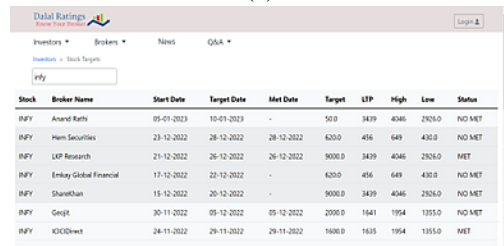
(a)



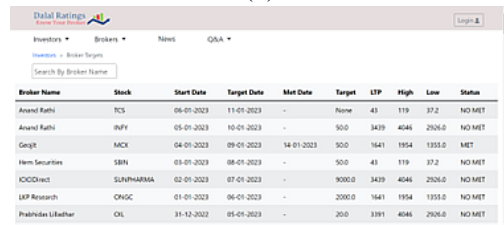
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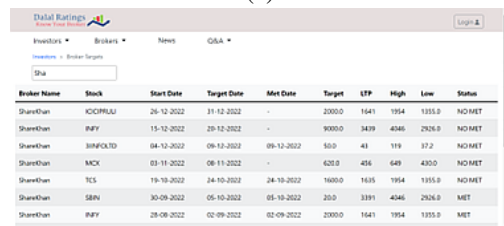
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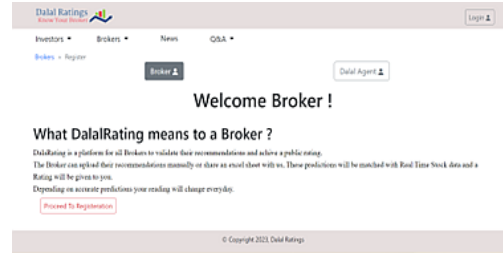


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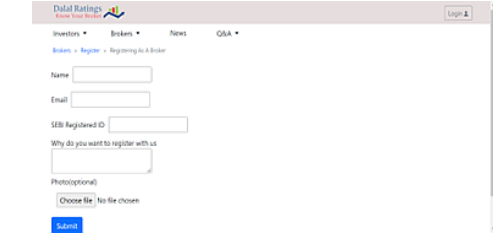


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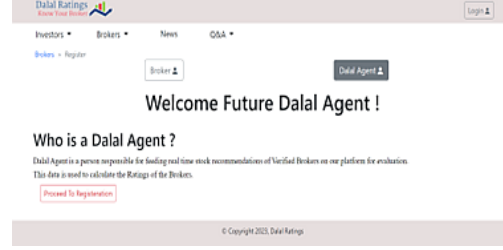
Fig. 5. Website page corresponding to: (a) Brokers ratings (b) Investors>Broker Ratings>See Recommendations (c) Investors>Stock Targets (d) Search Results (e) Investors>Broker Targets (f) Search Results for Broker Sharekhan



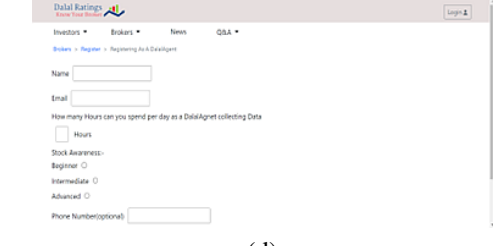
(a)



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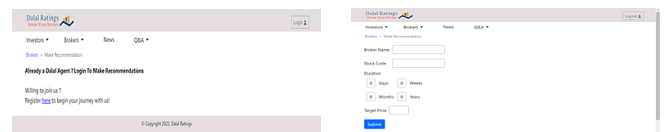


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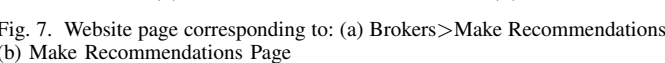


(d)

Fig. 6. Website page corresponding to: (a) Register page (b) Registration Form for Broker (c) Broker>Register (d) Registration Form for DalalAgent



(a)



(b)

Fig. 7. Website page corresponding to: (a) Brokers>Make Recommendations (b) Make Recommendations Page

sources. Each DalalAgent is given a username and password upon registration. They are responsible for uploading data periodically. This data can also be uploaded by the Brokers him/herself.

(g) News The user can navigate to the page shown in Fig. 8 by clicking on news in the navbar. This website also keeps the user updated with the latest financial news on the market. The user can further read the complete news article by clicking "

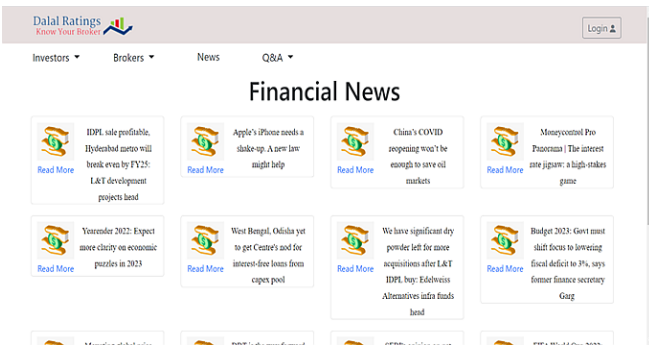


Fig. 8. Website page for News

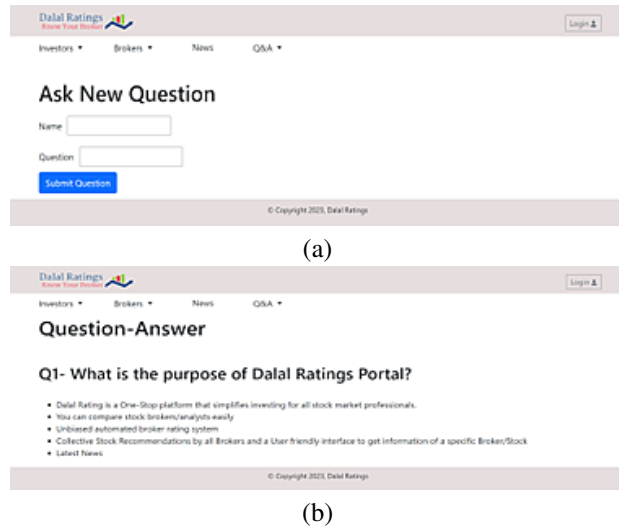


Fig. 9. Website page corresponding to: (a) Q&A>Report Issue (b) Q&A>FAQ

read more. The news is updated daily and taken from trusted and verified sources.

(h) Common Question/Answer Our proposed work also provides the feature to ask new questions, report issues, and view frequently asked questions. Fig. 9(a) shows the Report issue/Ask Question Page. Figure 9(b) shows the frequently asked questions (FAQ) page.

(i) Stock Forecasting Several methods related to deep learning are employed successfully in various real-world applications [19], [20]. Even, the prediction of stocks is done by using machine and deep learning methods [21]–[23]. Our proposed work also provides the feature to get predicted stock prices trained using LSTM Neural Networks with two different architectures and great accuracy directly in the user's mail. In this, the pre-processing steps involve tackling the missing values, if any, by using the imputation technique such as forward fill, then the hyperparameter tuning is done by using the Grid Search method. The user needs to enter the desired Stock Code and their email ID; they can expect the result to be shared within 1 minute. Fig. 10(a) shows the Stock Forecasting Page. Fig. 10(b) shows the confirmation message sent by the mail on the Stock Forecasting Page as soon as the user submits

the details. Fig. 10(c) shows the result email received by the user if the stock code entered was valid and trained on model 1. Fig. 10(d) shows the result email received by the user if the stock code entered was valid and trained on model 2. Fig. 10(e) shows the attached graph in an email received by the user indicating whether the stock code entered was valid and trained on model 1. Fig. 10(f) shows the result of the email received by the user that the wrong stock code was entered. Although the model was able to provide promising initial accuracy values, errors can still occur due to dynamic nature of stock market. The performance of the model varied across different time periods, such as during the abrupt shifts in the market trends. Thus, there is a possible scope for additional features, such as capturing short-term fluctuations or unexpected market trends.

V. COMPARATIVE ANALYSIS OF PROPOSED WORK WITH EXISTING TECHNOLOGY

Based on the study, the features of the existing systems, namely, the Money Control Website, Economic Times Markets Website, and NSE Official Website, are listed in Table II along with the features of proposed work. Since the internet is easily available to everyone, accessing our proposed website has become very easy for everyone, be it a stock market expert or just a beginner.

TABLE II
COMPARISON TABLE OF EXISTING AND PROPOSED WORK

System Name	Money Control [15]	Economic Times [14]	NSE [16]	Proposed Work
Application type	Web-based	Web-based	Web-based	Web-based
Cost of installing	Free	Free	Free	Free
Verify Recommendations	No	No	No	Yes
News	Yes	Yes	Yes	Yes
Collective Recommendations	Yes	Yes	No	Yes
Automatic Broker Rating System	No	No	No	Yes

VI. CONCLUSION AND FUTURE SCOPE

This paper explains the workings and importance of the Stock Market, the need for platforms like Dalal Rating, and how to use our proposed solution in the real world effectively. The Dalal Rating portal is a unique platform with multiple features that will help increase the efficiency of stock market professionals in the long run. The user-friendly interface makes it easy for everyone to utilize our portal to get effective outcomes. The stock prediction is also obtained from the LSTM Neural Networks. Shortly, we plan to automate the "Make Recommendation" feature by extracting information from Audio/Video recordings of various News channels, thus making our portal more secure and effective by processing huge amounts of data with less human effort and errors.

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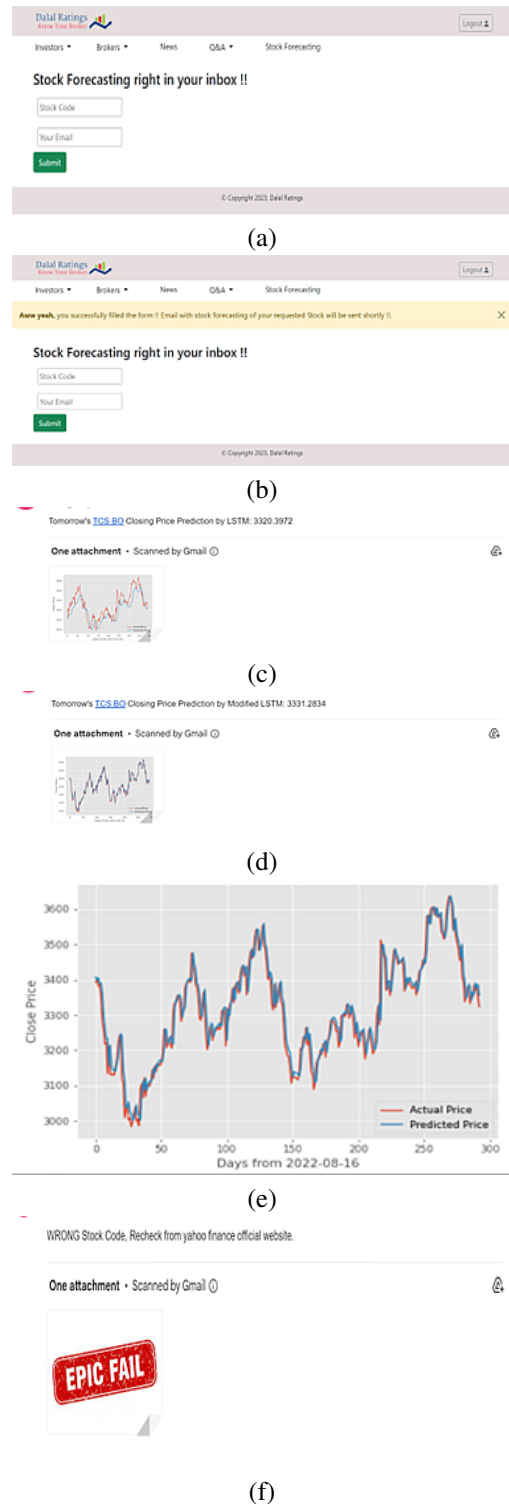


Fig. 10. Website page corresponding to: (a) Stock Forecasting (b) Email Confirmation (c) Email Response for valid Stock Code (Model 1) (d) Email Response for valid Stock Code (Model 2) (e) Graph attached in Email Response (f) Email Response for invalid Stock Code.

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